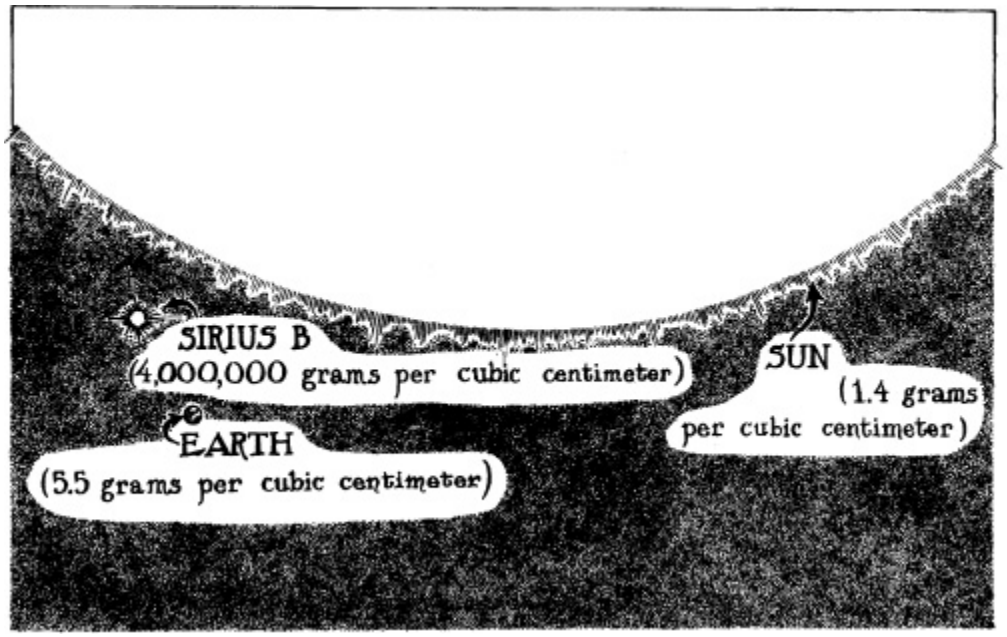
mountaintop above Pasadena, California.[2] “Professor Adams has killed two birds with one stone,” Eddington wrote; “he has carried out a new test of Einstein’s general theory of relativity and he has confirmed our suspicion that matter 2000 times denser than platinum is not only possible, but is actually present in the Universe.”



4.1 Comparison of the sizes and mean densities of the Sun, the Earth, and the white-dwarf star Sirius B, using modern values.

Further on in Eddington’s book, Chandrasekhar found a description of how the internal structure of a star, such as the Sun or Sirius B, is governed by the balance of internal pressure against gravitational squeeze. This squeeze/pressure balance can be understood (though this was not Eddington’s way) by analogy with squeezing a balloon in your hands (left half of Figure 4.2): The inward force of your squeezing hands is precisely counterbalanced by the outward force of the balloon’s air pressure—air pressure that is created by air molecules inside the balloon bombarding the balloon’s rubber wall.